Business Intelligence Assignment 2

Submitted By:
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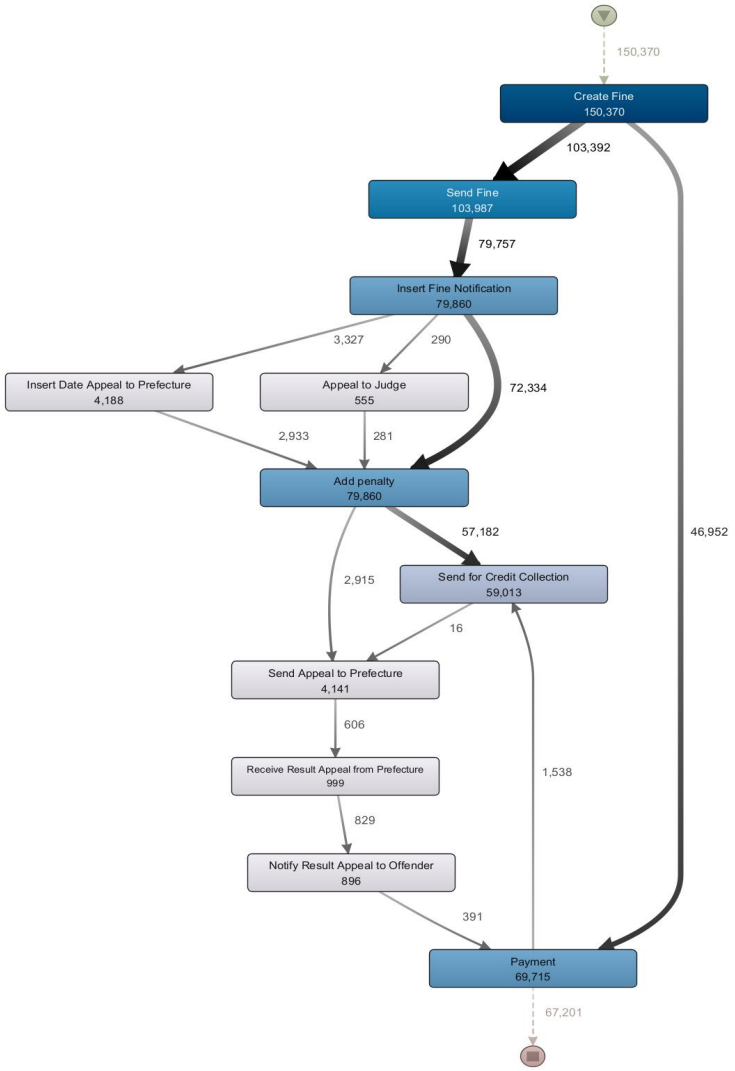
Task 1: Process Discovery - Loading dataset into disco

Event Log Selection:
Road Traffic Fines Management process:

- After loading the dataset, we got a process map.
- We observe the initial process map generated by Disco.
- It shows all activities and paths, which were too detailed.
- It was a bit complexed, so we decided to adjust the slider to adjust the map, making it easier to understand.



Task 1: Process Discovery - Adjusting the Slider (Phase 1)

Adjusting the Map:

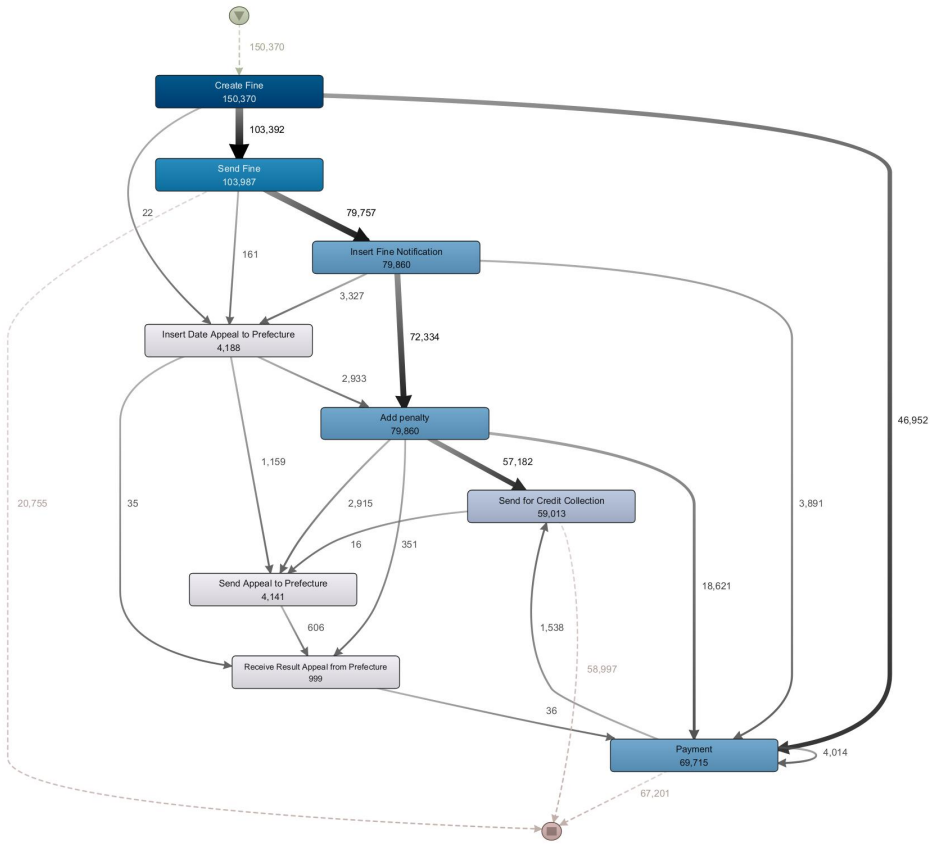
- Moved the Paths slider to around 50%.
- Moved the Activities slider to around 70%
- The process map appears to be more streamlined but still includes some complexity.

Key Activities:

- The main activities like Create Fine, Send Fine, Insert Fine Notification, Payment, are clearly visible.

- There are other activities related to appeals and notifications.

- There are still multiple paths, especially related to the appeals process and notifications.



Task 1: Process Discovery - Adjusting the Slider (Phase 2)

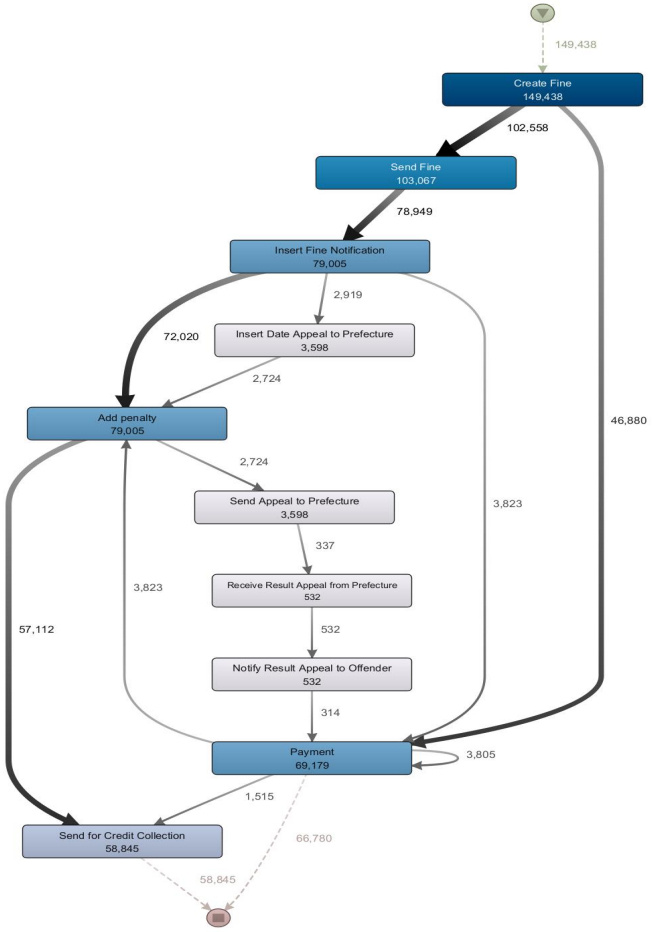
Adjusting the Map:

- Move the paths slider to around 40% and review the map.
- Applied a frequency filter to include cases that occur in more than 10% of the total cases.

Key Activities:

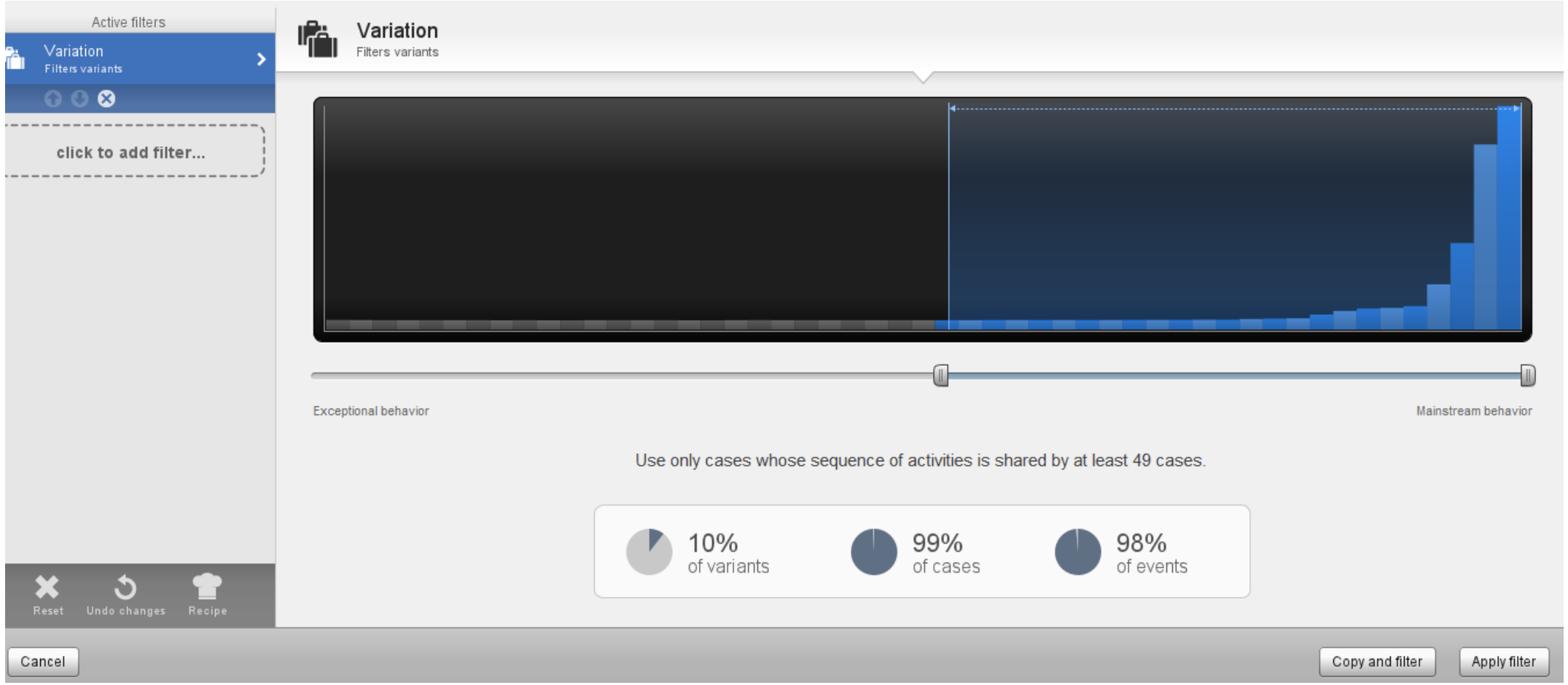
- The primary activities such as Create Fine, Send Fine, Insert Fine Notification, Payment and Add Penalty are again prominently displayed.

- The paths and activities are significantly reduced, focusing on the most frequent flows.



Task 1: Process Discovery - Adjusting the Slider (Phase 2)

Choosing 10% variants of total cases.



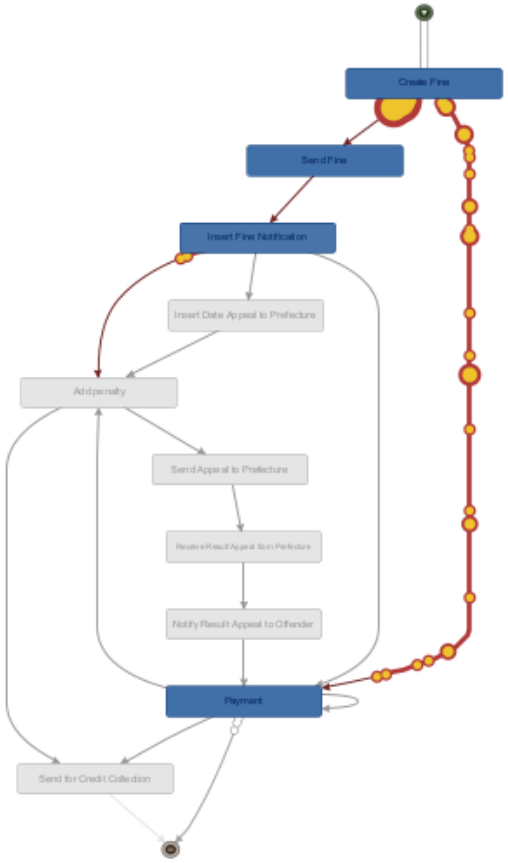
Task 1: Process Discovery - Adjusting the Slider (Phase 2)

Animation:
We used the animation to observe the process.
After creating fine, 3 processes are started:

1. Payment
2. Send Fine
3. Add penalty

We analyzed that:

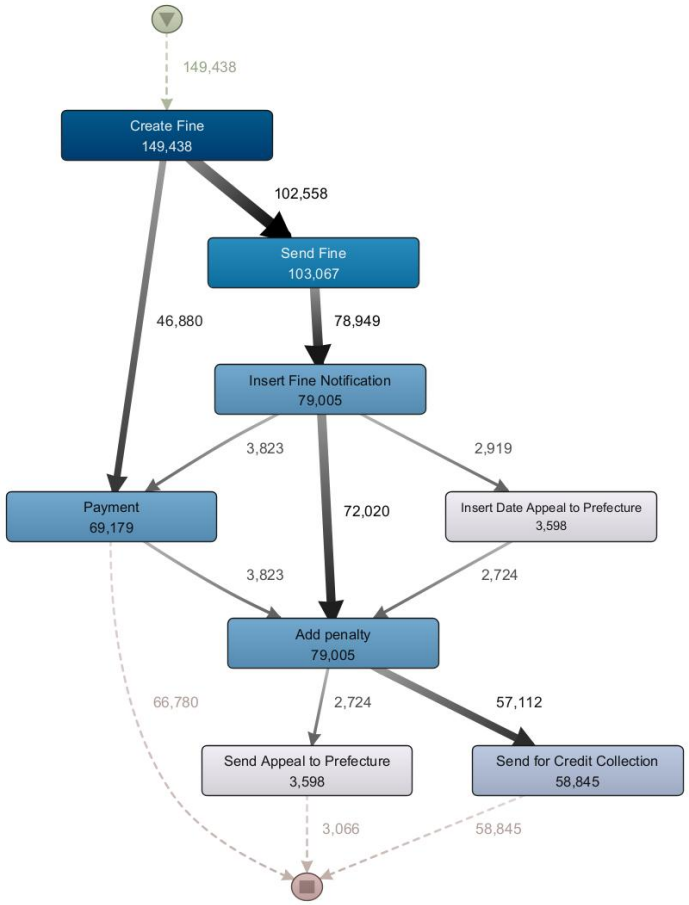
- After payment, the process should not go to “Add Penalty” again rather it should go directly to Credit Collection, review it and they should add penalty and send fine again.



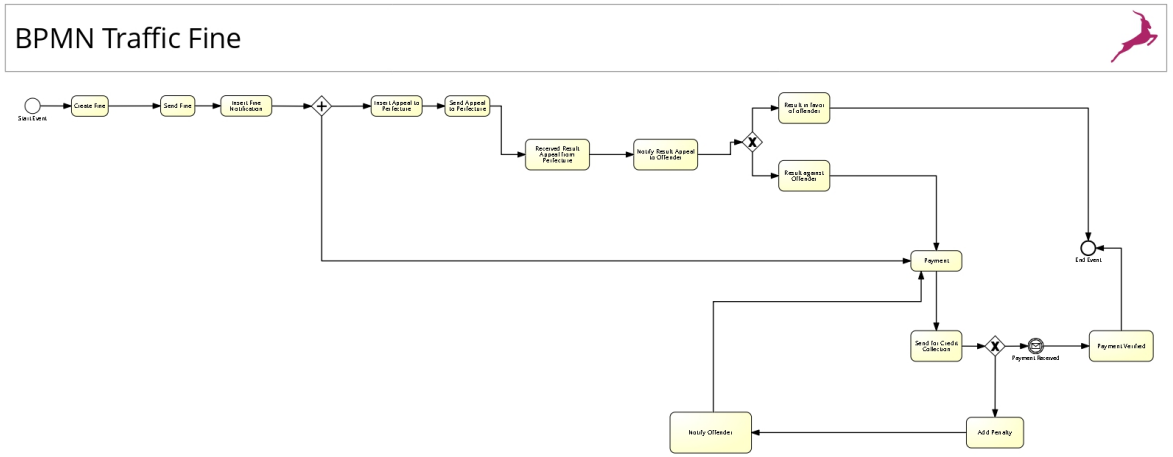
Task 1: Process Discovery - Adjusting the Slider (Phase 3)

- After analyzing the process map by adjusting paths and adding frequency filter, we still analyzed some processes which did not seem us fine. So, we tried to further reduce the activities to 30% and paths to 10%.
- We still see complexities like, after sending appeal to prefecture it was being directly halted, whereas there should be further activities to display the result of the appeal either it was canceled or not.

So, we created a BPMN model that gave us more control over the content.



Task 1: Process Discovery - BPMN Model



BPMN Model – Details Included

By analyzing the initial process map generated by disco, which was a bit detailed and complexed. We decided to include the activities that represent the main steps in the road traffic fine management process. The activities that were essential and not complicating the process. Also ensuring that not any process was missed. These activities include:

- Create Fine: Initiates the fine process.
- Send Fine: Sending the fine to the offender.
- Insert Fine Notification: Recording the fine notification in the system and sending notification to the offender.
- Insert Appeal to Prefecture: Logging the appeal if the offender contests the fine.
- Send Appeal to Prefecture: Sending the appeal to the authority for reconsidering fine.
- Receive Result Appeal from Prefecture: Receiving the outcome of the appeal.
- End Events: The model includes end events to signify the completion of the process, like:

Successful completion when payment is verified.

Termination if the process ends in favor of the offender without further action required.

BPMN Model – Details included

- Notify Result Appeal to Offender: Informing the offender of the appeal decision.
- Payment: Managing the payment of the fine.
- Send for Credit Collection: Sending fines for collection.
- Add Penalty: Adding penalties for late and unpaid fines.
- Payment Verified: Verifying that payment has been received.
- Decision Points: Key decision points that determine the flow of the process.

BPMN Model – Details Omitted

The sub-process that were not at right place, and complicating the model were omitted. For example:

- Adding penalty after inserting notification.
- Adding penalty directly after Payment.
- Adding penalty after inserting date appeal to prefecture.

These were omitted because they were causing a problem like after payment, the payment should be sent to collection department to verify and collect if payment is complete or not and then adding penalty accordingly.

Certain exceptions were not included like

- Appeal to Judge

Task 2: Process Conformance

1. Does it ever happen that a fine is paid before it is sent out?

Relevance:

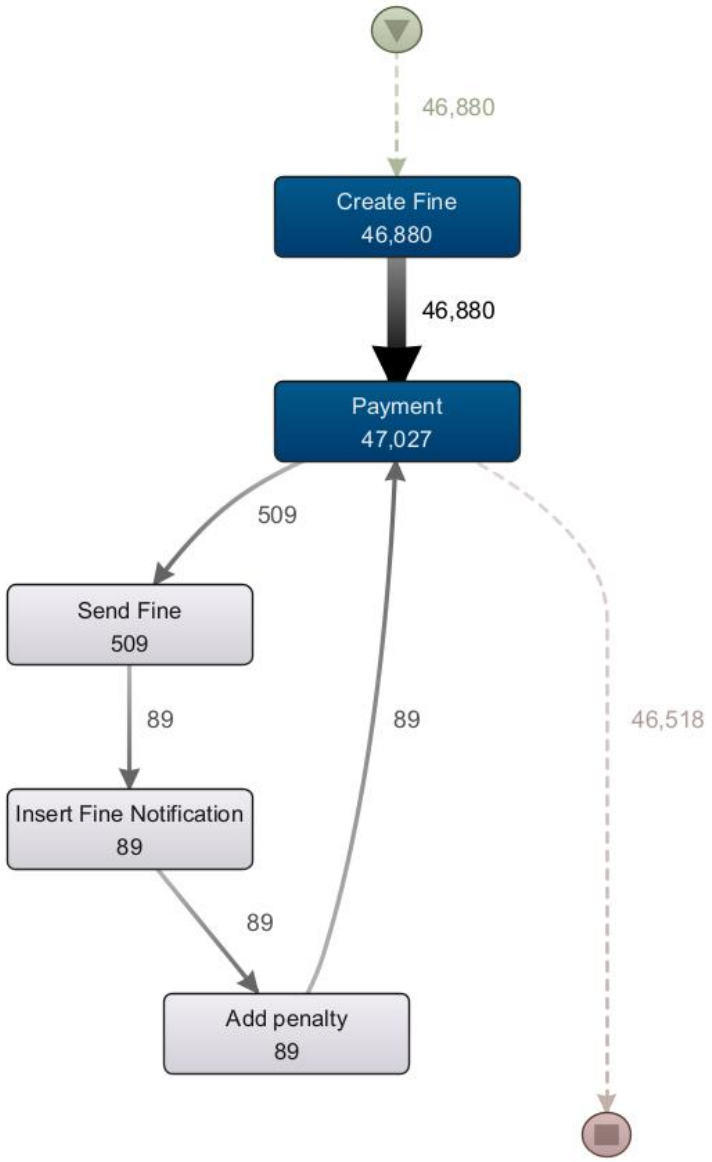
In traffic fine management process, a fine must be send before it is paid out. If payments are made without fine being issued, it depicts there is an issue in the process, which means there is financial discrepancies and lack of accountability. This suggests that there might be errors in record keeping or in the system. This helps to ensure financial integrity.

Investigation:

To investigate this, we applied follower filter with refrence case “Create Fine” directly followed by “Payment” on top 10% variants.

Result:

We received 46,880 such cases where the fine is being paid before it is sent out, so this shows a great issue in our case. We have also attached a process map for this issue.



Task 2: Process Conformance

2. Are there cases where a payment reminder is sent after the fine has already been paid?

Relevance:

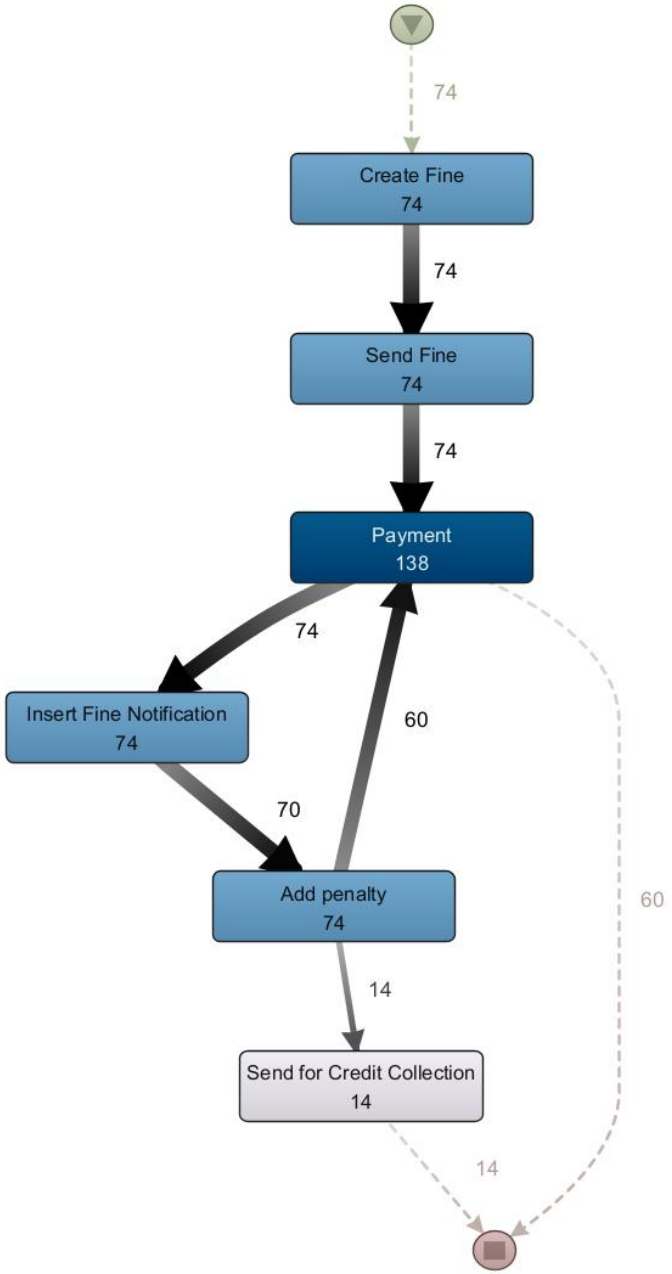
In this case, where a payment reminder is sent after the fine has been paid shows poor communication and process inefficiency, potentially causing confusion and dissatisfaction among recipients. It also indicates that there is some issue in the system and resources are wasted on unnecessary reminders.

Investigation:

We investigated this issue by applying follower filter with refrence case “Payment” directly followed by “Insert fine notification”. We also checked this by applying filter with refrence case “Payment” eventually followed by “Insert fine notification”. But this can also happen in some cases, where after payment a penalty is added and then notification is sent again, so we didn’t take this into account, because this is not an issue.

Result:

The process map after applying filter showed us that there are 74 such cases, where a reminder is still being sent even if the fine is already paid.



Task 2: Process Conformance

3. What is the typical outcome for cases that involve Send appeal to the Prefecture? How often do these appeals result in fines being re-assessed or cancelled?

Relevance:

Understanding the outcomes of appeals is important for assessing the fairness and effectiveness of the fine management process. If there are more cases who's fine is being cancelled after the appeal, this shows that there are issues with the initial fine assessment accuracy.

Investigation:

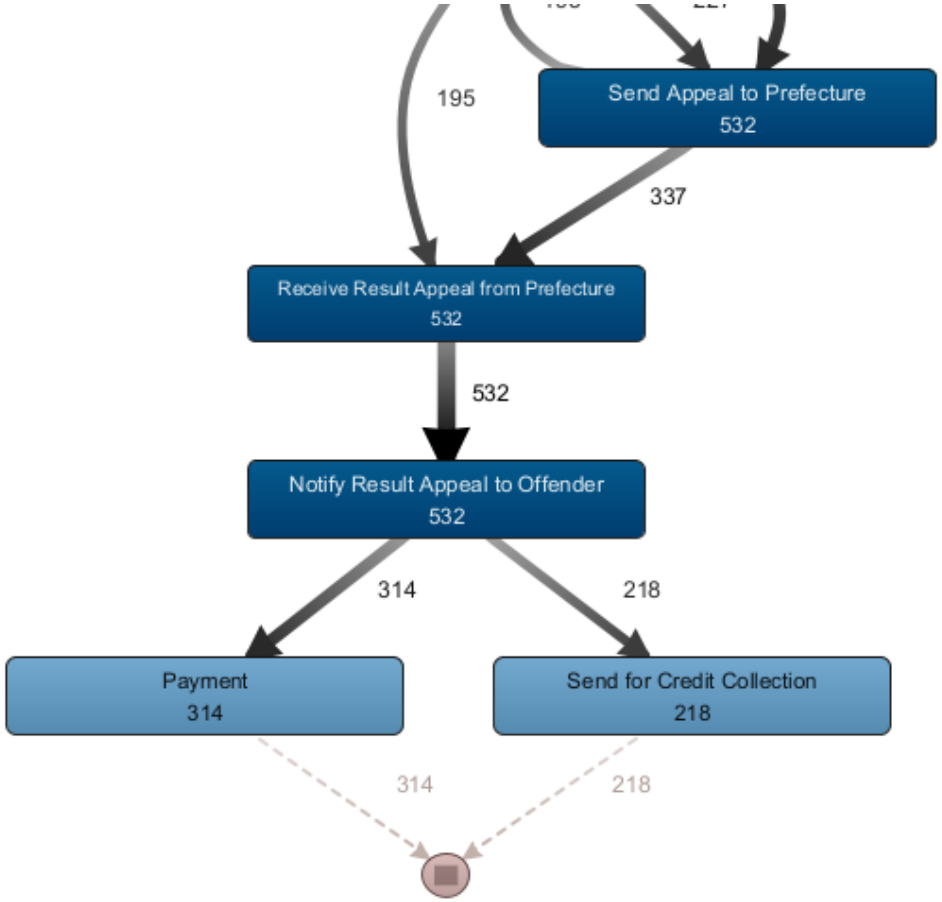
We investigated this by applying following filter with refrence case “Receive Result appeal from Prefecture” directly followed by “Notify result appeal to offender” and “ Payment”. We investigated this by counting the number of cases from the process map that got notified from the result, the number of cases which have done payment and the number of cases who's fine was cancelled.

Task 2: Process Conformance

3. What is the typical outcome for cases that involve Send appeal to the Prefecture? How often do these appeals result in fines being re-assessed or cancelled?

Result:
By investigating we got the result that there were total 532 cases who got notified from the result out of which 314 were those whose fine was re-assessed and they can be seen doing payment. On the other hand, the remaining 218 cases were those whose result was cancelled and can be seen going to send for credit collection department because they had nothing to pay, and their fine was cleared.

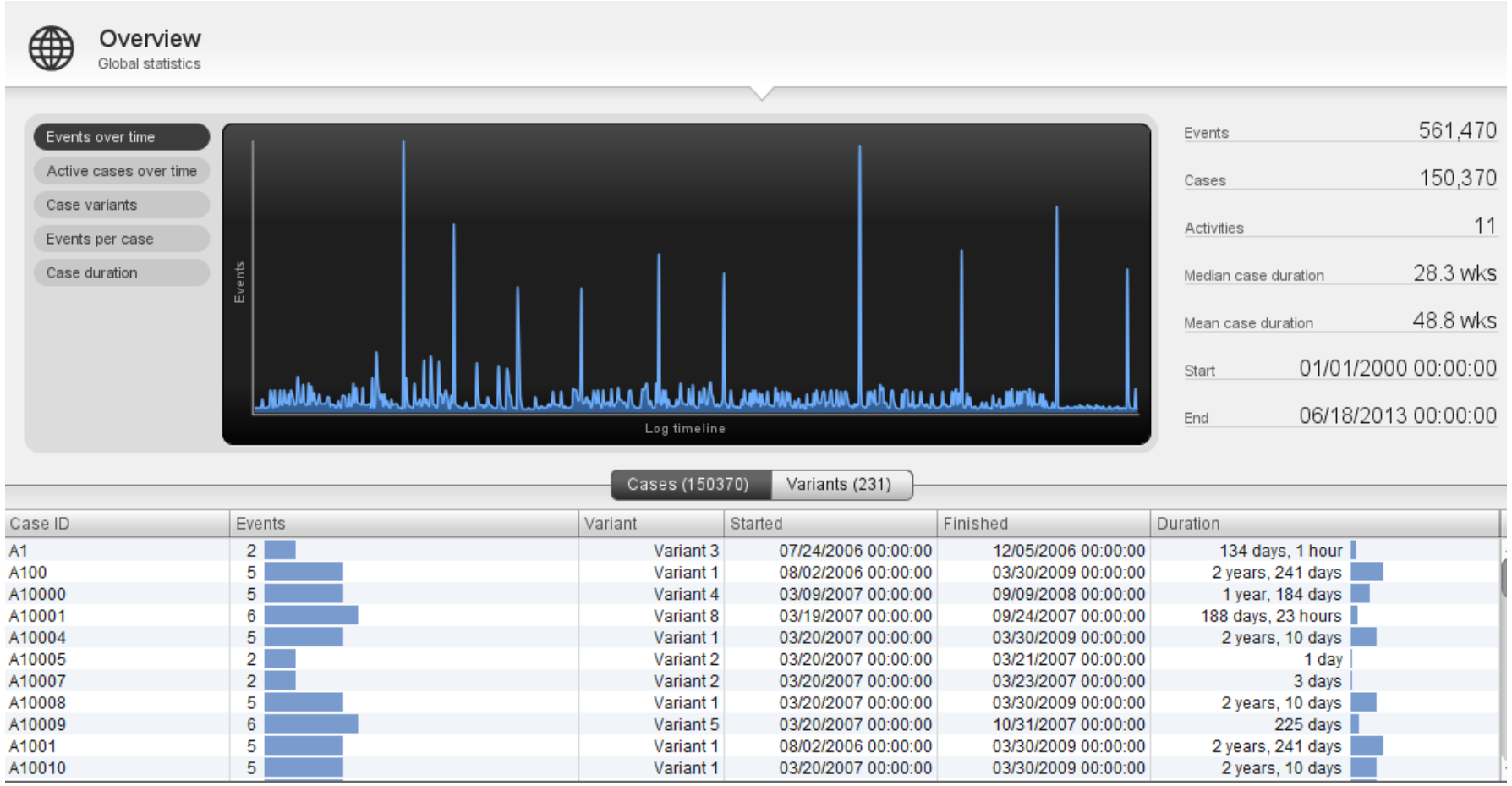
So, this mean there were cases who doesn't have fine but were accounted for payment.



Task 3: Time Perspective

1. How long do process instances take to execute and where are the main bottlenecks in the execution?

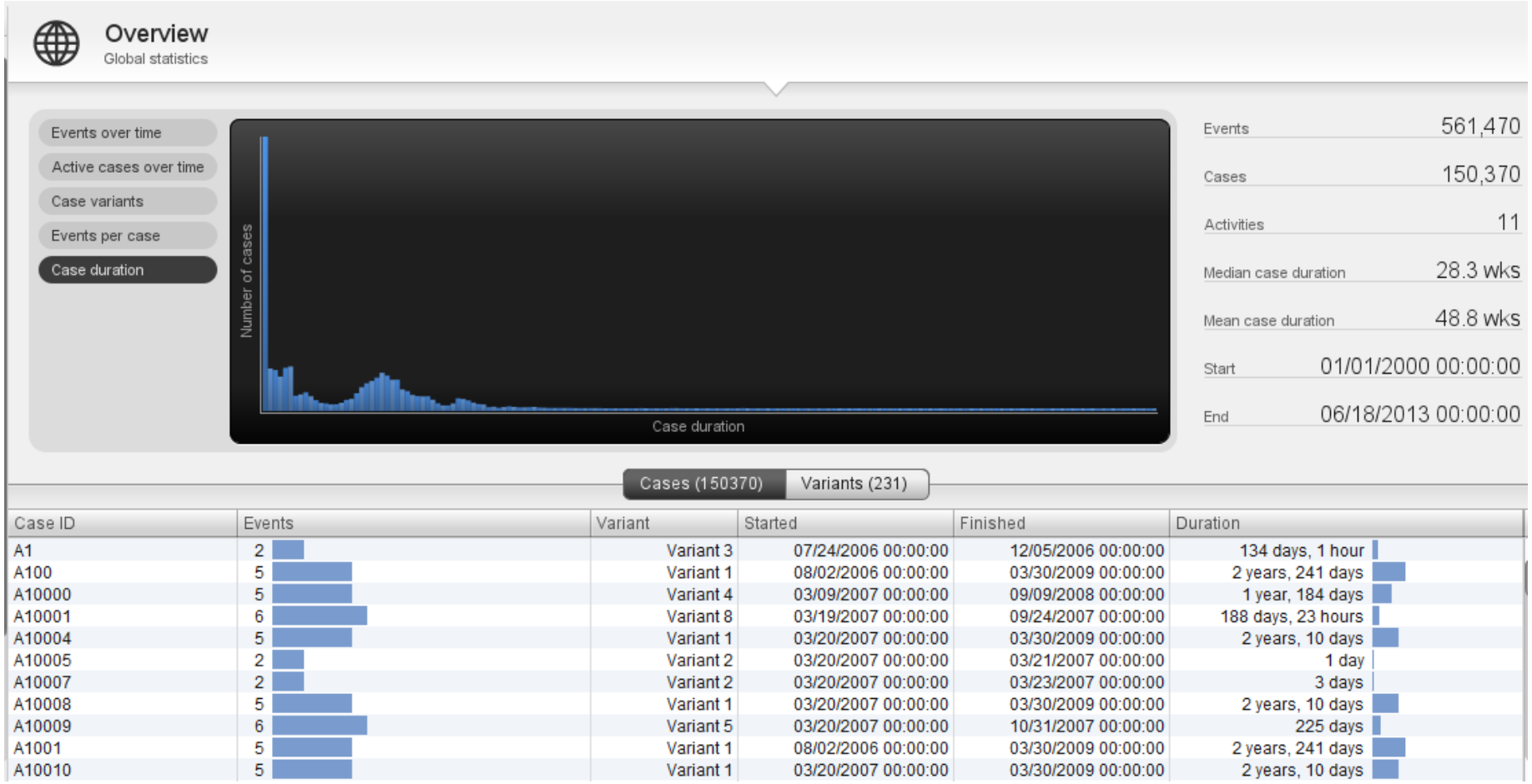
We went though the statistics view to have an overview of the process instances. This shows the events over time and there are 561,470 events and 150,370 cases with the median case duration is 28.3 weeks and mean case duration is 48.8 weeks.



Case ID	Events	Variant	Started	Finished	Duration
A1	2	Variant 3	07/24/2006 00:00:00	12/05/2006 00:00:00	134 days, 1 hour
A100	5	Variant 1	08/02/2006 00:00:00	03/30/2009 00:00:00	2 years, 241 days
A10000	5	Variant 4	03/09/2007 00:00:00	09/09/2008 00:00:00	1 year, 184 days
A10001	6	Variant 8	03/19/2007 00:00:00	09/24/2007 00:00:00	188 days, 23 hours
A10004	5	Variant 1	03/20/2007 00:00:00	03/30/2009 00:00:00	2 years, 10 days
A10005	2	Variant 2	03/20/2007 00:00:00	03/21/2007 00:00:00	1 day
A10007	2	Variant 2	03/20/2007 00:00:00	03/23/2007 00:00:00	3 days
A10008	5	Variant 1	03/20/2007 00:00:00	03/30/2009 00:00:00	2 years, 10 days
A10009	6	Variant 5	03/20/2007 00:00:00	10/31/2007 00:00:00	225 days
A1001	5	Variant 1	08/02/2006 00:00:00	03/30/2009 00:00:00	2 years, 241 days
A10010	5	Variant 1	03/20/2007 00:00:00	03/30/2009 00:00:00	2 years, 10 days

Task 3: Time Perspective

This is an overview of all case durations.



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A10001	6	Variant 8	03/19/2007 00:00:00	09/24/2007 00:00:00	188 days, 23 hours
A10004	5	Variant 1	03/20/2007 00:00:00	03/30/2009 00:00:00	2 years, 10 days
A10005	2	Variant 2	03/20/2007 00:00:00	03/21/2007 00:00:00	1 day
A10007	2	Variant 2	03/20/2007 00:00:00	03/23/2007 00:00:00	3 days
A10008	5	Variant 1	03/20/2007 00:00:00	03/30/2009 00:00:00	2 years, 10 days
A10009	6	Variant 5	03/20/2007 00:00:00	10/31/2007 00:00:00	225 days
A1001	5	Variant 1	08/02/2006 00:00:00	03/30/2009 00:00:00	2 years, 241 days
A10010	5	Variant 1	03/20/2007 00:00:00	03/30/2009 00:00:00	2 years, 10 days

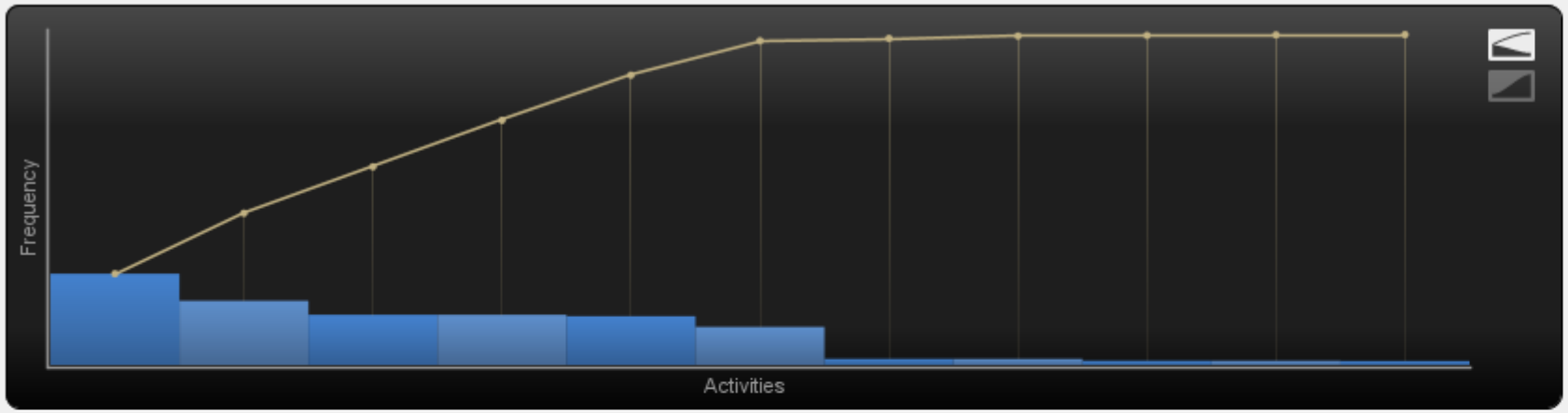
Task 3: Time Perspective

By analyzing the events and the case duration, we inferred that:

- The median represents the middle value of the case durations, this means half of the cases are completed in less than 28.3 weeks and the mean is the average duration of all cases..
- The mean is much higher than the median showing that there are some cases with very long durations.
- The histogram shows a significant peak on the left side, indicating those cases that are completed relatively quickly, within a short time.
- There is a long tail extending to the right, which represents cases that take a much longer time to complete.
- This long tail indicates the presence of outliers.
- These outliers are likely due to specific bottlenecks or inefficiencies in the process that cause certain cases to be delayed significantly.

Task 3: Time Perspective

After going through the distribution and identifying presence of outliers, We went to the statistics and go through the “Activity” overview to examine the frequencies of each activity.



Activities	11
Minimal frequency	555
Median frequency	59,013
Mean frequency	51,042.73
Maximal frequency	150,370
Frequency std. deviation	52,021.11

All activities (11) | First in case (1) | Last in case (7)

Activity	Frequency	Relative frequency
Create Fine	150,370	26.78 %
Send Fine	103,987	18.52 %
Insert Fine Notification	79,860	14.22 %
Add penalty	79,860	14.22 %
Payment	77,601	13.82 %
Send for Credit Collection	59,013	10.51 %
Insert Date Appeal to Prefecture	4,188	0.75 %
Send Appeal to Prefecture	4,141	0.74 %
Receive Result Appeal from Prefecture	999	0.18 %
Notify Result Appeal to Offender	896	0.16 %
Appeal to Judge	555	0.1 %

Task 3: Time Perspective

All activities (11) | First in case (1) | Last in case (7)

Activity	Frequency	Relative frequency
Create Fine	150,370	26.78 %

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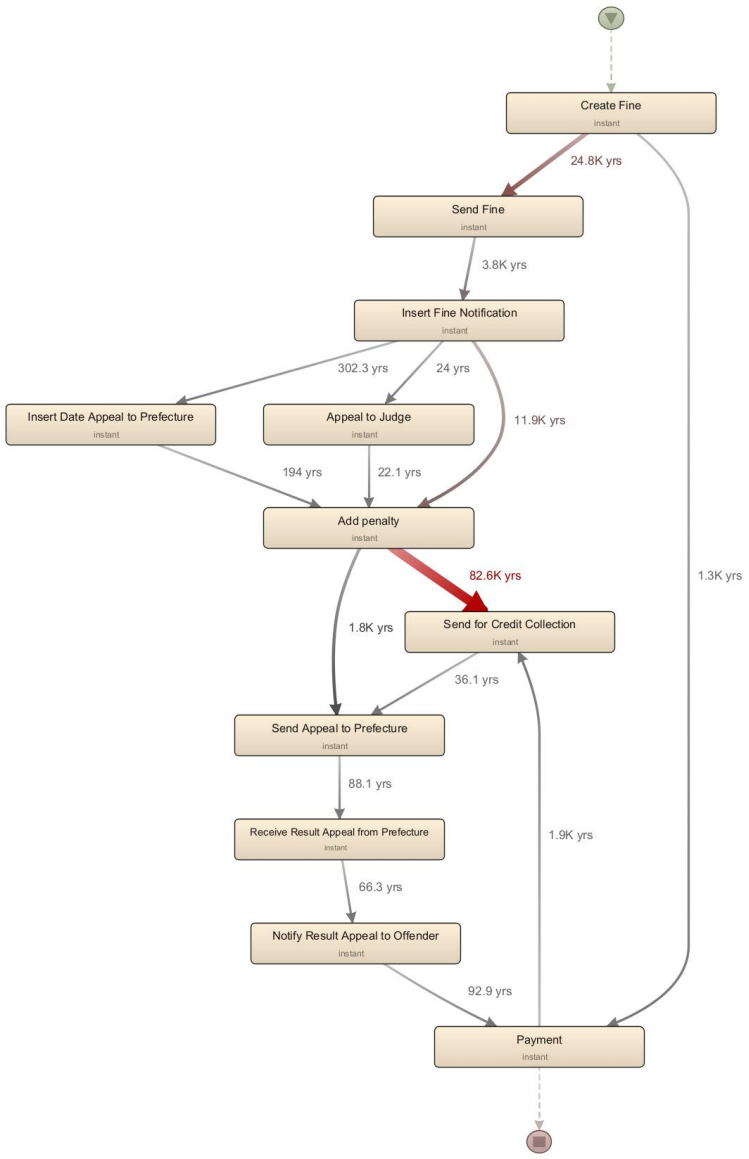
Send for Credit Collection	59,013	10.51 %
Send Appeal to Prefecture	4,141	0.74 %
Receive Result Appeal from Prefecture	999	0.18 %
Notify Result Appeal to Offender	896	0.16 %
Appeal to Judge	555	0.1 %

We analyzed that activities with lower frequency may represent potential bottlenecks if they involve complex processing or delays. For instance, "Appeal to Judge" has a very low frequency (243 times, 0.04%), indicating it is a rare event, likely involving special circumstances.

Task 3: Time Perspective

We go though the performance overview in the process map to see the average duration of each event.

- "Send for Credit Collection" with the frequency of 59,013 indicates that this activity occurs 59,013 times within the dataset. This high frequency suggests that sending fines to credit collection is a common action in the process.
- The long duration for a high-frequency activity indicates a major bottleneck. This step is likely causing delays in the overall process, leading to extended case durations.
- The activities with high frequency and long duration indicates a significant bottleneck in the traffic fine management process.



Task 3: Resource Perspective

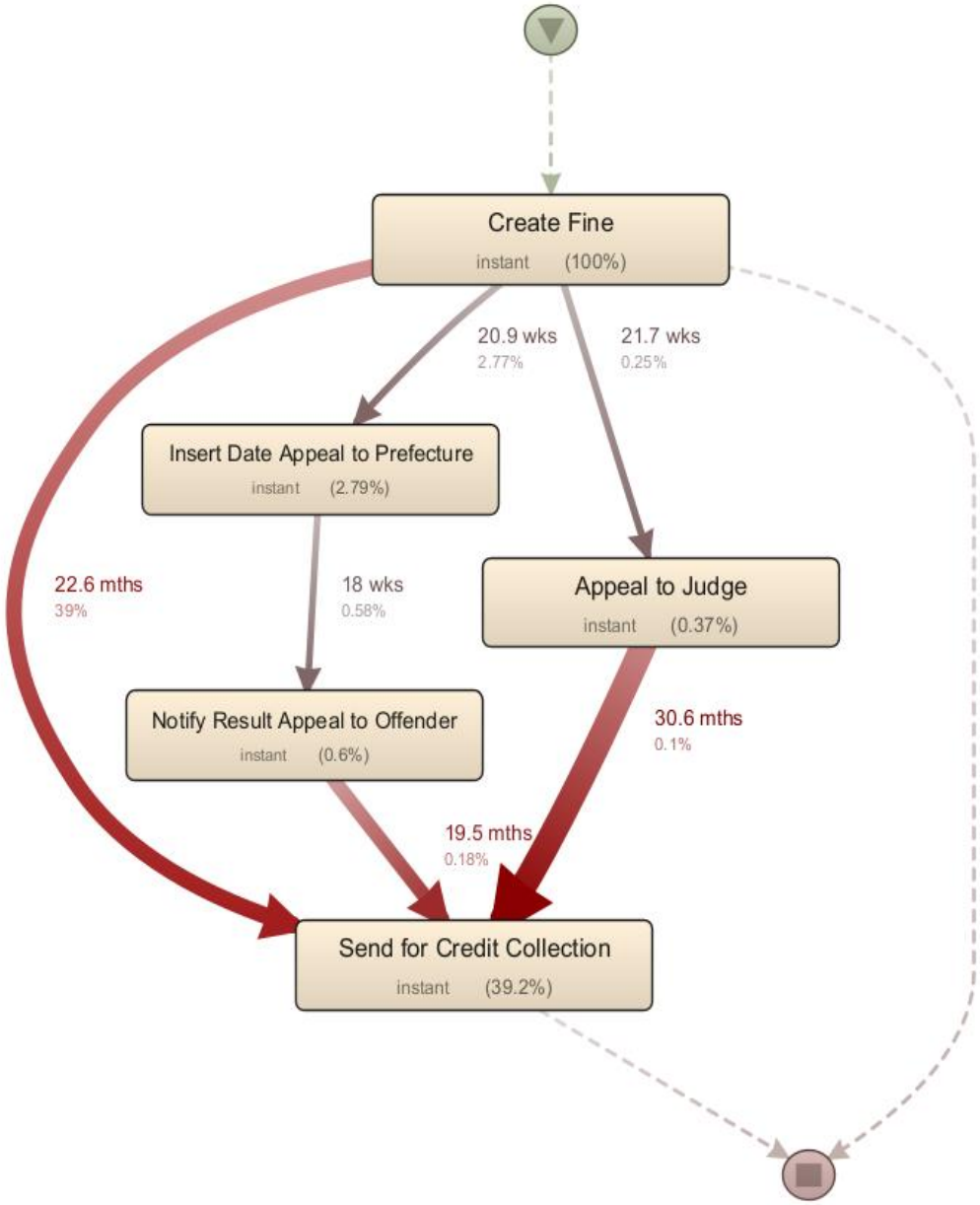
2. Is there specific resource behavior that affects the execution of an instance?

To examine the resource behavior, we first identified the resources, since there are no direct resources available in our dataset. So, we identified the activities including resources.

We applied the attribute filter for the resources, this allowed us to isolate the events handled by specific resources.

We got a process map and analyzed that;

- "Insert Date Appeal to Prefecture“ takes 22.6 months for 39% of the cases, which is significantly long. This means that the resources involved in this event are either too inefficient, or the workload is too much.



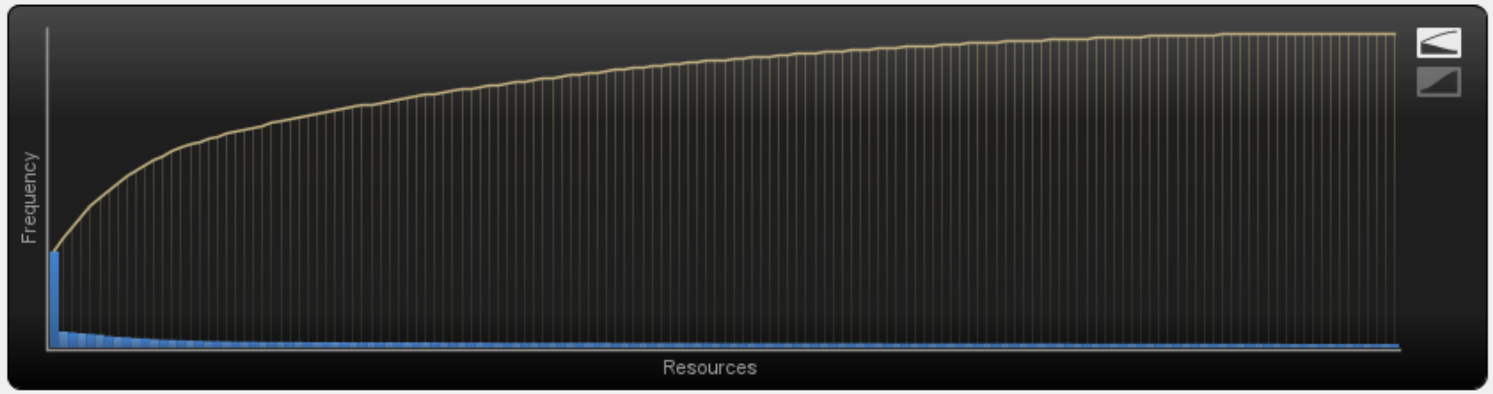
Task 3: Resource Perspective

- “Appeal to Judge” is taking 30.6 months for only 0.1% of cases. The long duration and low frequency might indicate that when it occurs, it is a highly time-consuming process, potentially due to judicial or administrative delays.

- "Send for Credit Collection" has almost 39.2% of cases going through this route instantly. This means particular resource or department handling credit collections; has a high efficiency in initiating collections.

We went though the resource statistics to further analyze the resources with respect to the frequency measures.

Task 3: Resource perspectives



Resources	149
Minimal frequency	2
Median frequency	501
Mean frequency	1,443.1
Maximal frequency	64,097
Frequency std. deviation	5,385.95

All resources (149) | First in case (147) | Last in case (149)

Resource	Frequency	Relative frequency
	64,097	29.81 %
538	8,608	4 %
550	7,935	3.69 %
541	7,356	3.42 %
537	6,931	3.22 %
559	6,429	2.99 %
557	5,436	2.53 %
536	4,821	2.24 %
49	4,589	2.13 %
561	4,043	1.88 %
558	3,779	1.76 %
40	3,183	1.48 %

Task 3: Resource perspectives

We analyzed that;

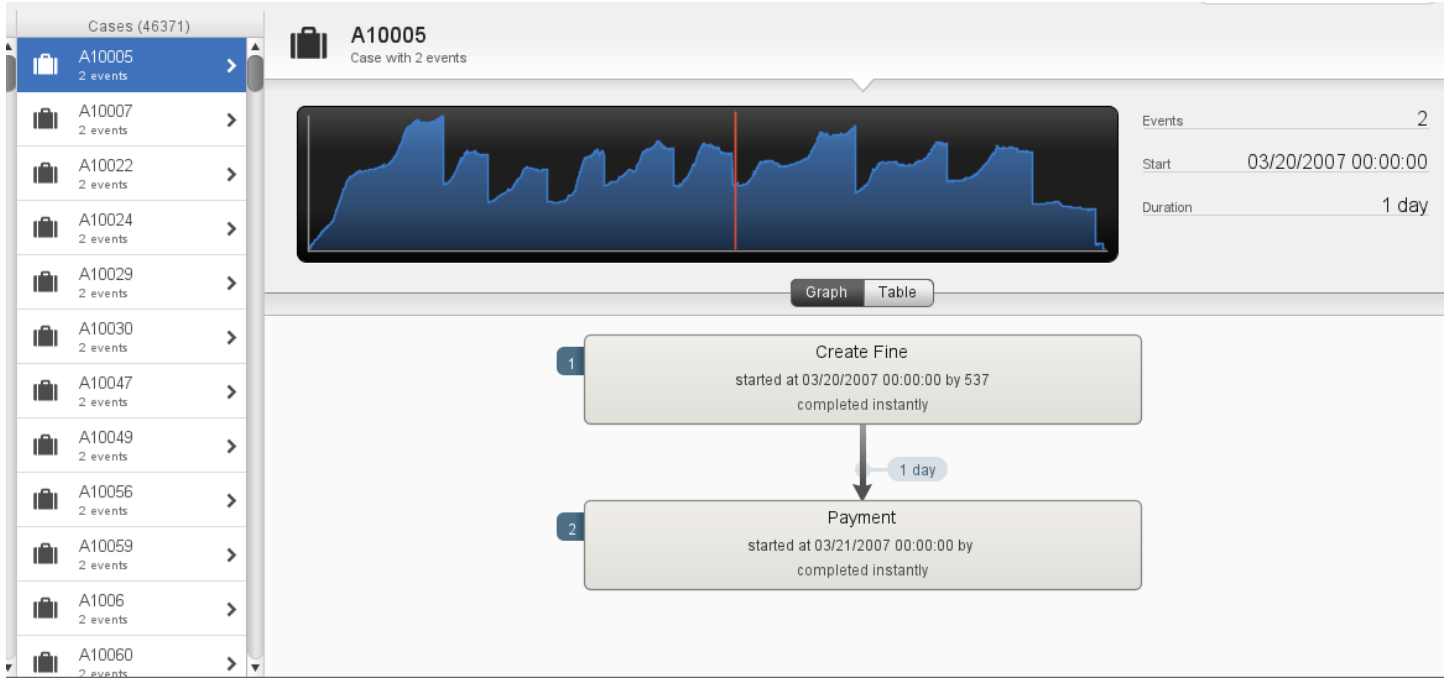
- The resource labeled 538 is the top resource with 64,097 occurrences in the events, which is about 29.81% of all the events handled by all resources. This means this resource is working more than the others.
- Following this, resources 550, 541, 537, etc., have handled fewer events, but still large amounts, ranging from 8,608 to 7,935 events, respectively.
- The least number of events handled by any resource is 2, indicating that this resource has been involved in only a few events.
- The median number of events per resource is 501, meaning half of the resources handle fewer than 501 events and the other half handle more.
- The maximum frequency of 64,097 events for a single resource shows that this resource has been overutilized.

Task 3: Data Perspective

3. What is the impact of data quality on process efficiency and outcomes?

To find out how data accuracy and completeness influence the process, we go through the statistics overview to analyze any potential gaps in the timeline that may indicate missing events or unusual data. We did not find out any missing or invalid data, However we examined how could data quality affects the execution of processes.

- The "Events over time" graph shows significant fluctuations with clear peaks and downs.
- We examined different cases to get an overview of how data quality affects process efficiency. 2 of which we have mentioned below:

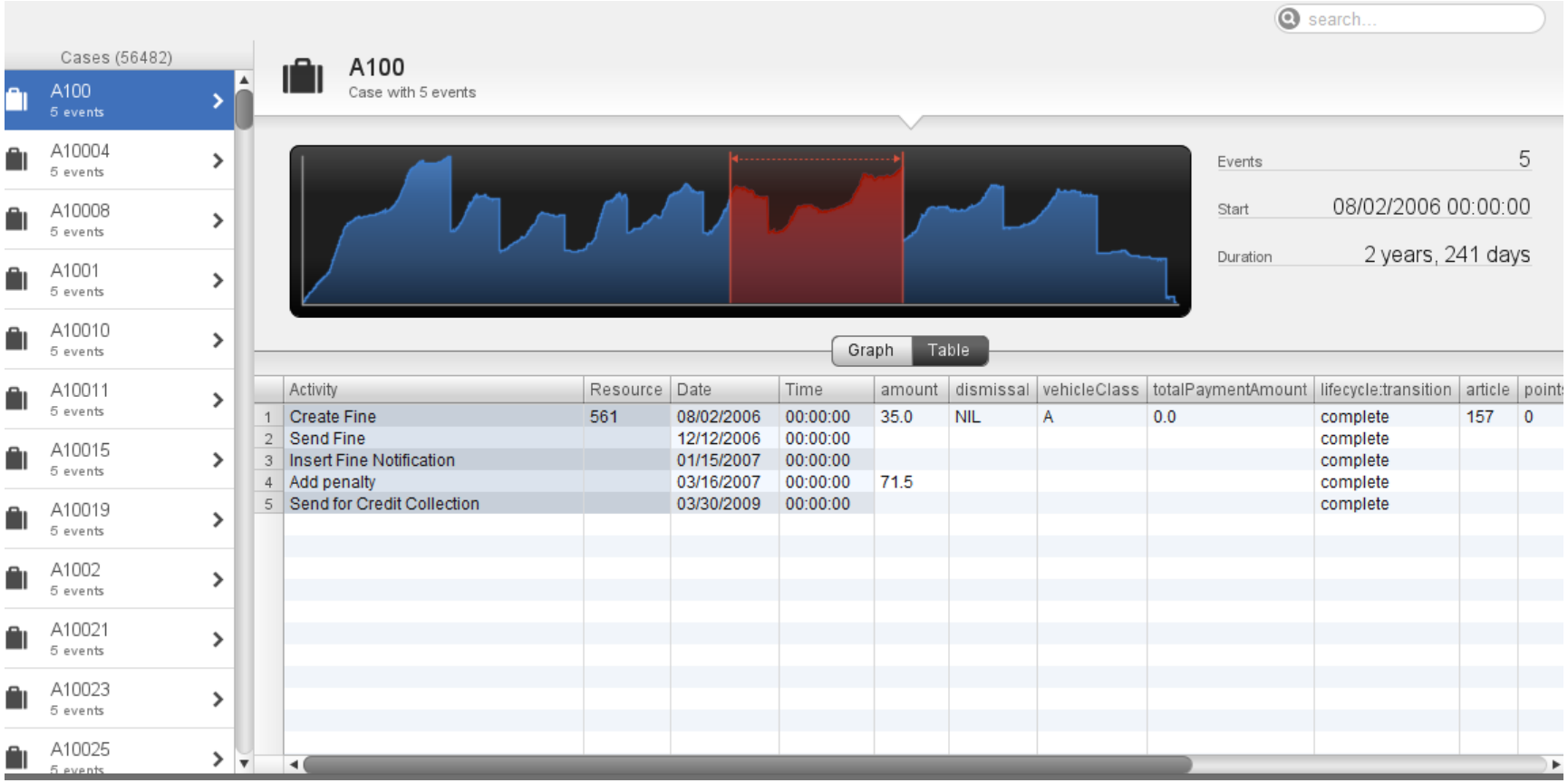


- This case comprises two events: "Create Fine" and "Payment completed within 1 day time
- The case data seems complete in terms of logging key events and their execution.
- The quick resolution of actions and the short case duration indicate an efficient process.

Task 3: Data Perspective

Task 3: Data Perspective

- The case consists of 5 events, with a total duration of 2 years and 241 days.
- The blue part of the graph indicated the activities or interactions that were performed.
- The red part indicates inactivity delay in the process event.
- The table helps verify that all events are logged with essential details, ensuring data completeness.
- The table shows the completeness of the activity's lifecycle, indicating that there was no data quality issue, but some other issues may be resource related, system errors or other challenges that caused the delay.



	Activity	Resource	Date	Time	amount	dismissal	vehicleClass	totalPaymentAmount	lifecycle:transition	article	point
1	Create Fine	561	08/02/2006	00:00:00	35.0	NIL	A	0.0	complete	157	0
2	Send Fine		12/12/2006	00:00:00					complete		
3	Insert Fine Notification		01/15/2007	00:00:00					complete		
4	Add penalty		03/16/2007	00:00:00	71.5				complete		
5	Send for Credit Collection		03/30/2009	00:00:00					complete		

Task 4: Further reflection

Which data or information are you missing that would enable you to gain additional insights into the process you are analyzing?

1. Detailed Resource Data
Data on how quickly and effectively each resource (employee or system) completes their assigned tasks.
2. Exception Handling
Detailed descriptions of different types of exceptions and how they are handled within the process.
3. Customer Feedback
Customer satisfaction scores or feedback related to the handling of fines.
4. Event Attributes
Reasons or comments associated with each event, providing more context for why certain actions were taken.
5. Detailed Financial Data
Information on the methods of payment used (e.g., online, in-person, bank transfer) and their impact on processing times.

Task 4: Further reflection

Do you see any clear improvement points in the process based on the analyses that you conducted?

Based on the information and analyses performed, particularly concerning the case details and the patterns observed in the event timelines, there are several clear improvement points that can be identified for the process

- **Address Process Delays**

The significant delays observed between some events, for example as the gap between "Add Penalty" and "Send for Credit Collection", suggest a need for process optimization. Streamlining these activities could reevaluate the steps and may reduce delays or long durations.

There should be more resources to shorten the response time and balancing the workload between each resource.

- **Implement Data Quality Controls**

Data Validation Checks could be introduced to automate the data validation process. As the data is entered, the quality of data should be checked.

By focusing on these improvement areas, the process can be made more efficient, responsive, and reliable